

CHAPTER 6

The Making of a Scientist

NCERT Exercise Solutions

- Read and Find Out (1) — 2 Q
- Picture question — 1 Q
- Read and Find Out (2) — 3 Q
- Think about it — 2 Q
- Talk about it — 2 Q

✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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⭐ Read and Find Out (1)

1 **How did a book become a turning point in Richard Ebright's life?**

✔ ANSWER

His mother got him a children's book, **The Travels of Monarch X**. It told how monarch butterflies migrate to Central America. It "opened the world of science" to him. At the end, readers were invited to tag butterflies for Dr Urquhart's research. Ebright began tagging and raising monarchs, and this led to all his later science projects.

2 **How did his mother help him?**

✔ ANSWER

- She encouraged his interest in learning.
- She took him on trips and bought him telescopes, microscopes, cameras, mounting materials and other equipment.
- She was his companion; they spent almost every evening learning together at the dining table, and she found learning work for him.
- She got him the book **The Travels of Monarch X** and wrote to Dr Urquhart, so he could tag butterflies for research.

❓ Picture question

1 **How is the monarch butterfly (top) different from the viceroy butterfly (bottom)?**

💡 SAMPLE ANSWER

Both are orange with black veins and white spots, so they look alike. The viceroy is a little smaller and has an extra **black line across its hind wings**, which the monarch does not have. (check the picture in your book)

⭐ Read and Find Out (2)

1 **What lesson does Ebright learn when he does not win anything at a science fair?**

✔ ANSWER

In the seventh grade his entry (slides of frog tissues under a microscope) won nothing. He realised the winners had tried to do **real experiments, not simply make a neat display**. He learnt that real science means doing experiments, and decided to do a real experiment the next year.

2 **What experiments and projects does he then undertake?**

✔ ANSWER

- **Eighth grade:** tried to find the cause of a viral disease that kills monarch caterpillars; raised caterpillars with beetles (no real results, but he won).
- **Next year:** tested the theory that viceroys copy monarchs; found that a starling would eat monarchs (first in zoology, third overall in the county fair).
- **High school (second year onwards):** studied the purpose of the twelve gold spots on a monarch pupa; built a device showing they produce a hormone needed for full development.
- **Senior year:** grew monarch wing cells in a culture; they developed into normal wing scales only when fed the gold-spot hormone.
- **At Harvard:** identified the hormone's chemical structure, and with James R. Wong made models and wrote a paper on a new theory of how cells read the blueprint of DNA.

3 **What are the qualities that go into the making of a scientist?**

✔ ANSWER

The text says: "Start with a **first-rate mind**, add **curiosity**, and mix in the **will to win for the right reasons**." Ebright also had hard work, determination, patience and the habit of doing real experiments.

🤔 Think about it

1 **How can one become a scientist, an economist, a historian... ? Does it simply involve reading many books on the subject? Does it involve observing, thinking and doing experiments?**

💡 **SAMPLE ANSWER**

Reading many books is not enough. Books give knowledge, but one becomes a scientist, economist or historian by **observing, thinking, asking questions and testing ideas**. Richard Ebright learnt this when his neat display of frog-tissue slides lost and the winners had done real experiments.

- **Curiosity:** a driving wish to know why and how things happen.
- **Observation:** Ebright watched monarchs through their whole life cycle.
- **Experiments:** a scientist tests ideas; an economist studies data; a historian checks sources.
- **Hard work and patience:** Ebright worked on the gold spots for years.

- **Guidance and will to win:** good teachers (Dr Urquhart, Mr Weiherer) and the wish to do one's best for the right reasons.

2 **You must have read about cells and DNA in your science books. Discuss Richard Ebright's work in the light of what you have studied. If you get an opportunity to work like Richard Ebright on projects and experiments, which field would you like to work on and why?**

💡 SAMPLE ANSWER

Ebright's work: The cell is the basic unit of life. DNA, found in the nucleus, carries the genetic information that controls heredity and decides the form and function of the cell; it is the "blueprint for life". Ebright found that the gold spots on a monarch pupa make a hormone needed for the wing cells to develop. Using X-ray photos of a hormone's structure, he gave a theory of how a cell "reads" its DNA. If correct, it can help us understand life processes and may lead to ways of preventing some cancers and other diseases.

My choice: I would like to work in **plant science (agriculture)**. Many farmers in India lose crops to pests and dry weather. I would like to do experiments to grow crops that need less water and fight pests naturally, so that farmers earn more and food is safe for everyone.

⭐ Talk about it

1 **Children everywhere wonder about the world around them. The questions they ask are the beginning of scientific inquiry. Given below are some questions that children in India have asked Professor Yash Pal and Dr Rahul Pal as reported in their book, Discovered Questions (NCERT, 2006).**

(i) What is DNA fingerprinting? What are its uses?

(ii) How do honeybees identify their own honeycombs?

(iii) Why does rain fall in drops?

Can you answer these questions? You will find Professor Yash Pal's and Dr Rahul Pal's answers (as given in Discovered Questions) on page 75.

✔ ANSWER

Answers as given by Professor Yash Pal and Dr Rahul Pal in the book:

(i) DNA carries genetic information unique to each living thing (except possibly an identical twin). When enzymes are added, DNA breaks at different places into fragments of different sizes; these move at different rates in an electric field and form a **unique pattern – the DNA fingerprint**. **Uses:** identifying parents (we inherit DNA from both), forensic science (a single hair or a spot of blood can prove guilt or innocence) and identifying bodies after accidents. It may one day be needed for ID cards, but it can also be misused (privacy).

(ii) Honeybees are very good at navigation: they use the **sun as a guide** and have a **good (pictorial) memory**. They tell the hive about food through a **dance** that shows the direction and distance of the food with respect to the sun. With this direction-finding and distance-reckoning, they easily find their way back home.

(iii) Rain forms when vapour condenses as air cools below the **dew point**. The vapour in a cloud cannot condense all at once into one big pool: pockets of air rise and cool separately, so separate **droplets** form. Most raindrops start as tiny ice crystals that gather moisture as they fall and melt into drops lower down (in cold places they reach the ground as snowflakes).

2 **You also must have wondered about certain things around you. Share these questions**

with your class, and try and answer them.

💡 SAMPLE ANSWER

- **Why is the sky blue?** Air scatters blue light from the sun much more than red light, so blue light comes to our eyes from all over the sky.
- **Why do onions make us cry?** Cutting an onion releases a gas that reacts with the water in our eyes and irritates them, so tears flow to wash it away.
- **Why do stars twinkle?** Starlight passes through moving layers of air with different densities, so it bends again and again and seems to flicker.

🎉 **Chapter 6 done! Revise the answers once more tomorrow.**